

Full Stack Development with MERN

Project Documentation

1. Introduction

- **Project Title:** Flightfinder : Navigating your air travel

options Team Members:

Team Size: 4

Team Leader:

- **Palla Harika**– Full Stack Development (Backend APIs, Database Integration, Project Integration) & Frontend Support & Testing **Team Members:**
- **Nikhitha Nadigatla**-Documentation & Presentation Support
- **Sravanthi Killi**– Frontend Development (UI Components & Design)
- **Kanakasri K S S N**-Project Deployment

2. Project Overview

- **Purpose:**

The purpose of this project is to develop a full-stack web application where it provide practical, job-oriented skills.Helps students gain hands-on experience beyond classroom theory.

Here user can register and log in,enroll in courses,a learning materials

- **Features:**



User Features:

- User Registration & Login
- Available trending courses
- Available explore course
- Afret login user can access learning materials

Admin Features:

- Admin Login
- Add new courses
- Edit new courses
- Delete available coourses

3. Architecture

Frontend (Client – React)

- Built using React.js
- Component-based architecture
- React Router for navigation
- Axios for API calls
- CSS for styling
- Runs on: <http://localhost:3000>

Frontend communicates with backend using REST APIs.

Backend (Server – Node.js + Express)

- Built using Node.js and Express.js
- REST API architecture
- Password hashing using bcrypt
- CORS enabled
- JSON request handling
- Runs on: <http://localhost:5173>

Database (MongoDB)

MongoDB used as NoSQL database.

Collections:

- Users
- Flights
- Registrations

Schemas created using Mongoose.

4. Setup Instructions

• Prerequisites:

- Node.js (v16 or above)
- MongoDB (Local or Atlas)
- VS Code

Installation:

Step 1: :<https://drive.google.com/drive/folders/1tDDOZqnWSgRxxXw20gyimiAPJI7WNRaz>

Open the link extract files in it.

Step 2: Backend Setup

```
cd backend
```

```
npm install
```

```
npm start
```

Backend runs on:

```
http://localhost:6001
```

Step 3: Frontend Setup

```
cdfrontend
```

```
npm install
```

```
npm start
```

Frontend runs on:

```
http://localhost:3000
```

5. Folder Structure

- **Client:**

client/

|

|— src/

| |— components/

| |— pages/

| |— styles/

| |— App.js/

| |— index.js

- **Server:**

server/

|

|— schemas.js

|— index.js

|— package.js

on

6. Running the Application

1. Start MongoDB
2. Start Backend:
npm install
3. Start Frontend:
npm start

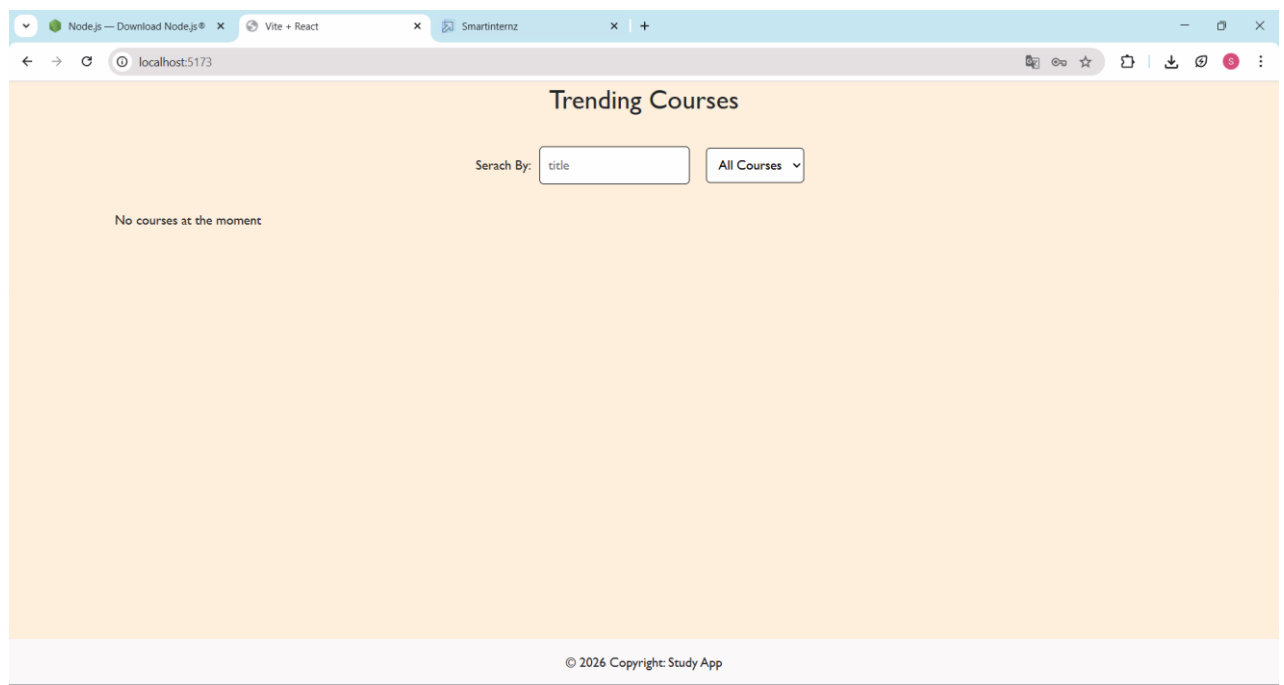
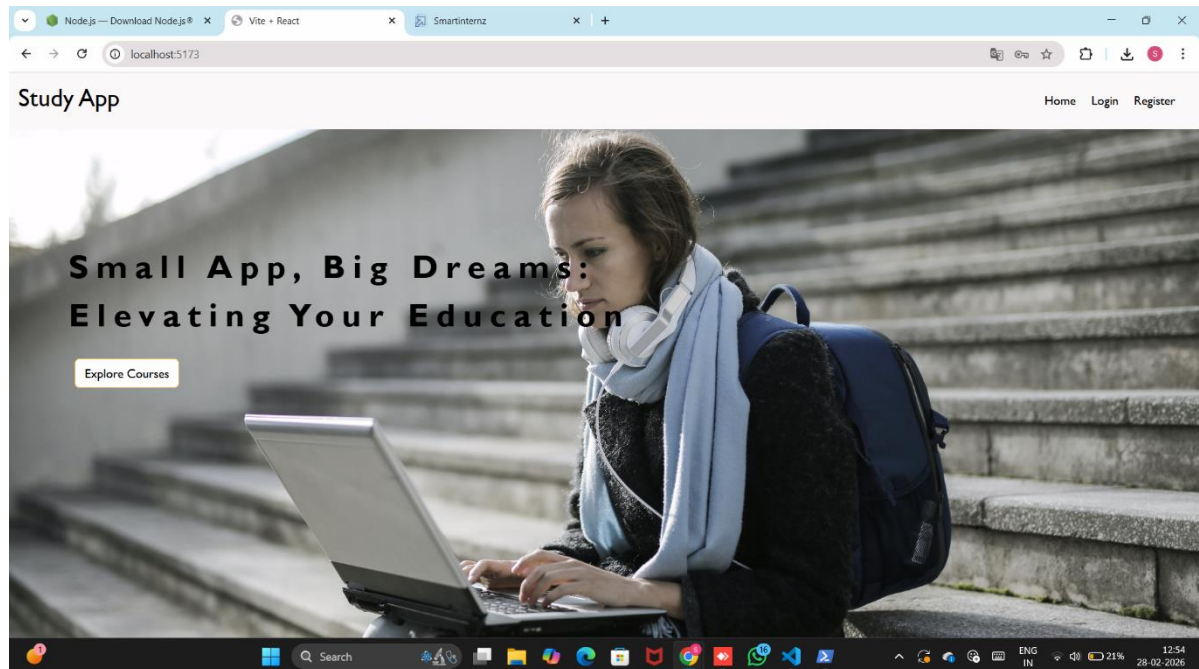
Open:

- Frontend → <http://localhost:3000>
- Backend → <http://localhost:6001>

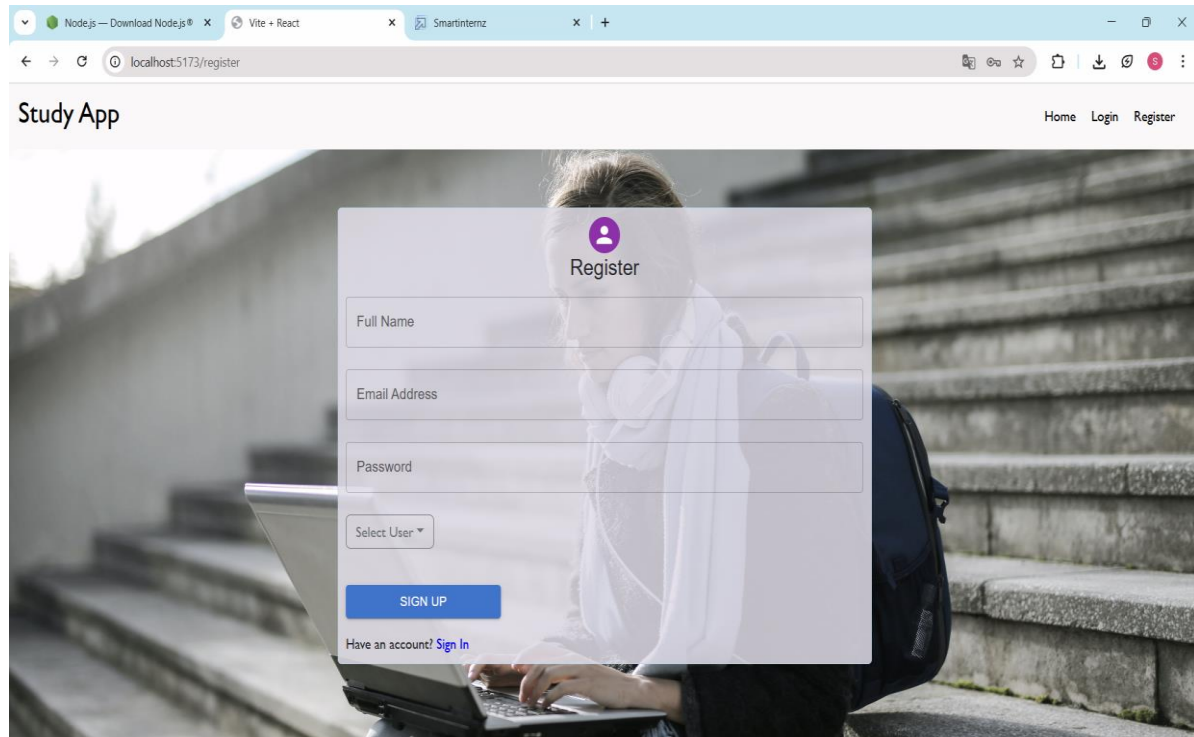
6.User Interface

- The UI includes:

Home Page



Registration Page



Study App

Home Login Register

Register

Full Name

Email Address

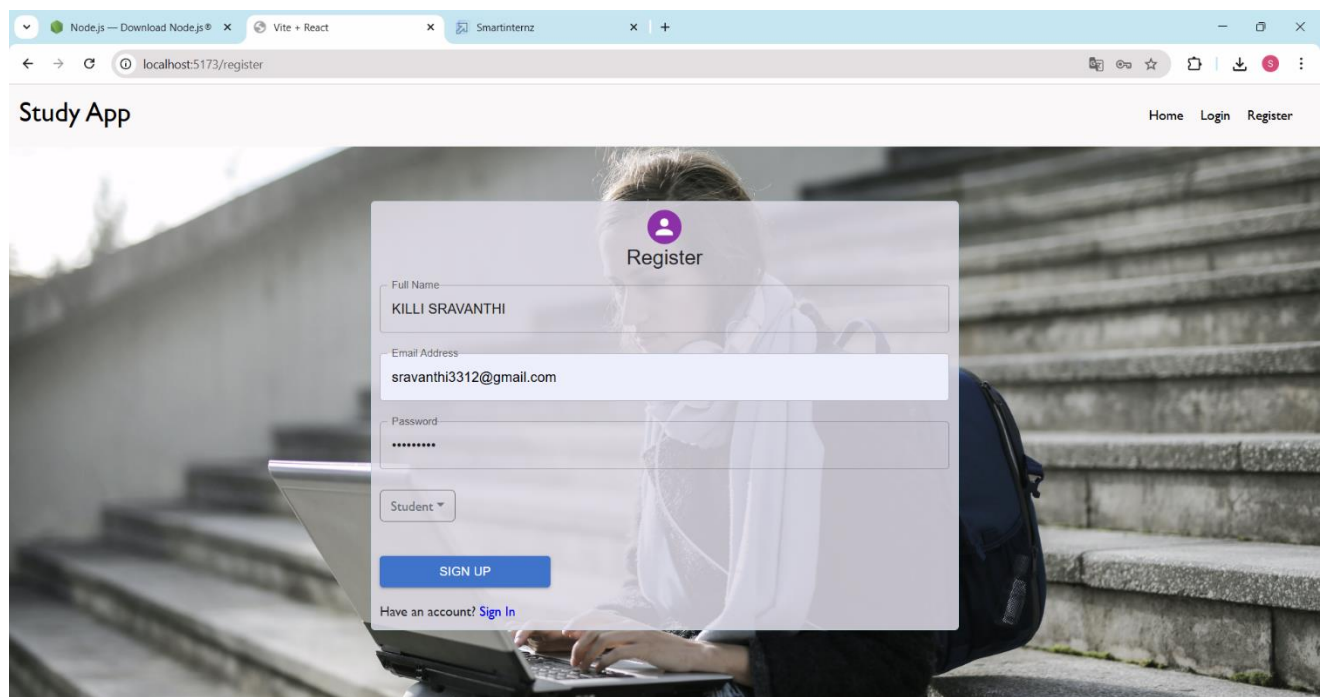
Password

Select User ▼

SIGN UP

Have an account? [Sign In](#)

Login Page



Study App

Home Login Register

Register

Full Name
KILLI SRAVANTHI

Email Address
sravanthi3312@gmail.com

Password

Student ▼

SIGN UP

Have an account? [Sign In](#)

1. Testing

- Manual testing performed for all APIs

Verified:

- Registration
- Login
- Add Courses
- View Registrations

Tested using:

Browser

2. Known Issues

- No email notifications
- Basic admin authentication
- UI responsiveness can be improved

3. Future Enhancements

- Add live class feature using WebRTC or third-party APIs.
- Recorded lecture upload system.
- Real-time doubt clarification.
- Timed mock tests.
- Automatic result generation.

